Wang Yuliang

Tel: (+65) 8506 8584 | Email: YULIANG005@e.ntu.edu.sg | LinkedIn: www.linkedin.com/in/yuliang-wang-b16153350
Github: https://github.com/love-ovo
Address: 757 Jurong West Street 74, Singapore 640757

EDUCATION:

Nanyang Technological University (NTU) **Master of Science** ***2026.08-2027.06***

- **Major:** Maritime Studies
- **Courses:** Ship Operations,Brokering and Chartering • Marine Insurance and Claims • Maritime Logistics • Sustainable and Green Finance • Sustainability and Digitalisation

Guangzhou Maritime University (GZMTU) **Bachelor of Science** ***2022.09-2026.06***

- **Major:** Maritime Affairs Management
- **GPA:** 89/100 (Ranking: 1/78)
- **Courses**: Systems Engineering • Systems Modelling and Simulation • Maritime Traffic Engineering• Advanced Python Programming • Container Transportation Practice

RESEARCH:

Leader, **5th International Supply Chain Modeling and Design Competition: Optimized Low-Carbon Supply Chain Network with Multi-Model Integration for Spare Parts Distribution** ***2025.02-2025.04***

- Optimized Company F's supply chain network by correcting distribution imbalances and inefficient inventory via an integrated low-carbon optimization model for parts and spares.
- Standardized geographical data with the Haversine algorithm, unifying distance metrics across five business datasets and completing unit conversion and data cleansing.
- Designed a five-node supply network with optimal coverage using a set covering model, formulating a mixed-integer program and solving it in Gurobi.
- Built a differentiated safety stock model by combining ABC-XYZ classification with service-level sensitivity analysis and minimizing annualized inventory cost.
- Verified solution robustness with discrete-event simulation (SimPy), testing order fulfillment rates and inventory stability.

Leader, **Mathematical Contest in Modeling (MCM): Agroecological Transition Driven by Multiscale Coupled Models: Resilience Optimization and Policy Levers (Meritorious Winner)** ***2025.01***

- Established dynamic equations using Logistic growth models, Lotka-Volterra equations, and Holling response functions, simulated long-term ecological evolution incorporating pesticide application and seasonal factors, and evaluated system stability through Lyapunov exponents.
- Developed Python-based ecosystem models to quantify ecological impacts of chemical pesticides, identified system chaos after 5 pesticide applications, measured 12% soil health degradation from herbicides, and observed 85% pest resurgence from insecticides.
- Designed restoration models integrating pollinator networks and decomposer functions, formulated a "3+2" strategy (3-year pesticide restriction plus 2-year bat habitat investment) using fuzzy robust control theory, achieved 18% farm bankruptcy rate reduction and 34% ecosystem stability improvement within 10 years.

Leader, **National Mathematical Modeling Contest: Application of Sampling Inspection and Optimization Models in Parts Quality Control (Third Prize in Guangdong Province)** ***2024.09-2024.10***

- Processed data in Python for normalization and dimensionless standardization to support model building and solving.
- Used sequential sampling inspection and Monte Carlo simulation to determine minimum sample sizes under different confidence and error requirements.
- Designed an optimal testing scheme that reduced inspection frequency and costs while maintaining accuracy.
- Built a cost-benefit model covering inspection costs, defect rates, and dismantling expenses.
- Optimized testing and handling strategies to balance quality risks and production costs.

Leader, **National Mathematical Modeling Contest: Optimization Model for Automated Pricing and Replenishment of Vegetable Products (Guangdong Provincial Excellence Award)** ***2023.09-2023.10***

- Researched automated pricing and replenishment strategies for supermarket vegetables to optimize supply-demand matching, increase profits, and reduce waste and inventory risks.
- Collected 2020-2023 sales data (retail/wholesale prices, spoilage rates), cleaned anomalies using Excel and SPSSPRO, and performed correlation/cluster analysis.
- Applied Pearson correlation to study cross-category sales relationships, identified bestsellers via clustering, built an ARIMA model to forecast weekly demand, and optimized restocking and pricing in an automated decision system.

Main Member, **2023 Nanfengchuang Research China: Impact of Marine Engineering Construction on Fishermen and Transformation under Maritime Power Strategy (National Third Prize)** ***2023.07-2023.10***

- Conducted in-depth interviews and questionnaire surveys with 50 fishermen in Huilai County, and collected data on economic status, social life quality, and offshore wind power development.
- Performed data coding, cleaning, and multivariate statistical analysis using SPSS PRO to ensure data quality and validity.
- Applied KMO test, factor analysis, and correlation coefficient analysis to identify key factors influencing fishermen's attitudes toward offshore wind power and livelihood transition.
- Used MATLAB for dimensionality reduction and feature extraction to optimize data structure and analytical accuracy.
- Implemented principal component analysis (PCA) and K-means clustering to assess fishermen's attitudes toward offshore wind power and explore potential pathways for fisheries transition.

INTERNSHIP:

Overseas OBD Intern, **Worldwide Logistics Group Co., Ltd.** ***2026.03-2026.04***

- Issued AWB, commercial invoices, and packing lists, ensured data accuracy, and complied with IATA standards.
- Managed airline bookings, coordinated carriers and ground agents, and optimized time-sensitive shipments.
- Assisted in preparing customs documents, verified HS codes, trade terms, and regulations for general and dangerous goods.
- Coordinated airline bookings and cargo operations for time-sensitive shipments.

Shipping Data Analysis Intern, **Guangzhou Yihai Shipping Co., Ltd.** ***2025.07-2025.08***

- Participated in shipping data visualization and analysis projects, supported route density identification, track analysis, and wind farm site selection.
- Conducted route identification, density analysis, and image processing based on AIS vessel trajectory data, processed large-scale trajectory data with Python and GIS, and enhanced visualization and decision support.
- Drafted shipping channel and wind farm site selection analysis reports, compiled key parameters, optimized presentation, and delivered project milestones.

Project Assistant Intern, **Guangzhou Xinhai Maritime Technology Services Co., Ltd.** ***2023.09-2024.10***

- Analyzed navigational safety impacts from operational activities, including the natural environment, port conditions, traffic patterns, and safety operation requirements.
- Proposed risk mitigation measures including safety warning signs, enhanced vessel communications, and guard boat deployment to reduce navigation risks.
- Participated in project evaluations and safety planning for drilling platform operations, subsea pipeline repairs, submarine cable maintenance/laying, and vessel towing operations.
- Organized expert review meetings to collect professional opinions on waterway navigation safety and compiled comprehensive safety assurance plans.

PUBLICATION:

First Author, Prediction of Shenzhen Port cargo throughput based on the nonlinear optimized GM-Markov model. *China Water Transport*, 25(3), 15-17.

First Author, Research on inland river crew team building under the new development pattern: Analysis based on Porter's five forces model. *Pearl River Water Transport*, 2024(10), 101-103.

Second Author, Improved GM-Markov model based on CPSO algorithm Port Cargo Throughput Forecasting, *Journal of Chongqing Jiaotong University (Natural Sciences)*

Second Author, Large-Scale Sampling Model and Optimization Algorithm Based on Dynamic Programming and Monte Carlo Simulation, The 5th International Conference on Multi-modal Information Analytics (MMIA)

Second Author, Research on Offshore Wind Farm Site Selection Optimization Based on Clustering Algorithm and Image Processing Technology, Journal of Guangzhou Maritime University

EXCHANGE EXPERIENCE:

6th Excellent University Students Summer Research Camp, University of Macau ***2025.07***

- Attended lectures on *Ocean Carbon Sequestration, Forecasting Models, and Carbon Removal Technologies*, enhancing understanding of marine environmental modeling and data-driven ocean observation.
- Explored marine pollution monitoring and early-warning systems through laboratory visits and academic exchanges, improving knowledge of marine environmental assessment methods.

5th International Summer School on Deep-0cean, Southern University of Science and Technology ***2026.07***

- Explored advancements in *Ocean AI, Digital Ocean Technologies, Deep-Sea Exploration, Blue Economy* through lectures, seminars, and industry visits.
- Built an interdisciplinary perspective on the role of AI, digital technologies, and marine innovation in advancing ocean science and sustainable ocean development.

STUDENT WORK:

Class Monitor, **Class 222, School of Navigation** ***2022.09-2024.09***

- Managed daily operations for a class of 40 students and processed various data.
- Organized 10+ events, including class meetings, college sports day opening ceremonies, and campus galas.

HONOURS & AWARDS:

WIFFA Scholarship	***2026.01***
Third-Class Scholarship, Excellence in Science and Innovation	***2025.06***
First-Class Scholarship, Excellence in Public Communications	***2025.06***
National Scholarship	***2024.12***
Outstanding Student Cadre	***2024.06***
Third Prize, "Yunchou Weiwo Cup" 19th National College Student Transportation Science and Technology Competition	***2024.04***
Second Prize, "Challenge Cup" 14th College Student Innovation and Entrepreneurship Competition	***2024.01***

LANGUAGE & SKILLS:

- **English:** IELTS: 6
- **Computer:** Proficient in MATLAB, Python, MS Office, LaTeX, Tableu, SQL and SPSS
- **Certificate:** NCRE (National Computer Rank Examination) - Level 2